



Question Bank with Answers

Chapter 4 — Web and Media Design

Exam-style practice questions

PROGRAMMING AND ARTIFICIAL INTELLIGENCE | SEMESTER 1

85 Questions Across 5 Sections

Fill in the blank • True/False • Matching • Multiple choice • Short answer • Analysis

QUESTION BANK

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Every question is reproduced exactly as written, grouped by lesson. Each question is tagged by format (fill in the blank, true/false, matching, multiple choice, short answer, analysis, or extended response) with its answer shown directly beneath in a green box.

REFERENCE • CHAPTER 4 QUESTION BANK

Chapter 4 — Quick Revision

4-1

TV = wide reach; radio = sound and can be received almost anywhere; newspaper = printed text/images; website = diverse and updatable; SNS = rapid sharing and two-way interaction.

4-2

Persona = who the typical user is. Wireframe = where information goes and its priority. CRAP = Contrast, Repetition, Alignment, Proximity. UCD = design from the user's perspective.

4-3

Qualitative = why. Quantitative = how much. PV = page views; bounce rate = visits ending after the first page; CVR = visits reaching a target action. Heuristic evaluation = experts use rules of thumb.

4-4

PDCA = Plan → Do → Check → Act, repeated. Design thinking = Empathize → Define → Ideate → Prototype → Test. A/B testing compares two options using real users and data.

4-1 • CHAPTER 4 QUESTION BANK

4-1

Types and Characteristics of Media

Worked Example → Try → Exam-Style [6 Marks] → Think as an Engineer → New Context → Reflect & Challenge → Exercise.

TRUE / FALSE

Q1

A website can present diverse information such as text, images, and video in combination. (True or False?)

ANSWER

True.

TRUE / FALSE

Q2

TV is a medium that allows two-way interaction with users. (True or False?)

ANSWER

False — TV is one-way.

TRUE / FALSE

Q3

SNS is highly effective at spreading information rapidly. (True or False?)

ANSWER

True.

TRUE / FALSE

Q4

A website normally can be viewed even without an Internet connection. (True or False?)

ANSWER

False — it normally requires an Internet connection unless cached.

MATCHING

Q5

Match: (a) detailed information viewable at any time; (b) users spread information by word of mouth; (c) inform many people at once with video. Choices: A TV, B Website, C SNS.

ANSWER

- (a) → B Website
- (b) → C SNS
- (c) → A TV

SHORT ANSWER

Q6

Name two strengths of websites.

ANSWER

Any two: diverse information; updatable; accessible from almost anywhere; two-way interaction; browser use without a dedicated app.

SHORT ANSWER

Q7

Name two constraints of websites.

ANSWER

Any two: normally needs Internet; users must actively access information; loading/display speed may vary.

MULTIPLE CHOICE

Q8

Which is NOT an appropriate website characteristic?

- A. Updated at any time.
- B. Automatically reaches users without active access.
- C. Combines text, images and video.
- D. Forms/comments allow interaction.

ANSWER

B.

MATCHING

Q9

Choose A TV, B Website, C SNS, or D Email: (a) detailed product information at any time; (b) word-of-mouth spreading; (c) video to many people at once; (d) direct campaign to customers.

ANSWER

- (a) → B
- (b) → C
- (c) → A
- (d) → D

6-MARK

Q10 6 marks

Explain why an organization might use a website, TV and SNS together rather than only one to promote a new service. Refer to reach, depth and word-of-mouth.

ANSWER

TV gives wide reach; the website gives detailed information at any time; SNS lets users share and spread the message.

THINK AS AN ENGINEER

Q11

A public library wants to promote a free reading program. Survey ten classmates/family members about printed posters, Facebook, WhatsApp and local radio. Consider a family with no home Internet, a student and the librarian. Recommend a media plan with two reasons.

ANSWER

Use survey evidence. Start from the audience hardest to reach and choose the medium that already reaches them. Combining media by purpose is often appropriate.

NEW CONTEXT**Q12**

A hospital wants to announce new opening hours and let patients ask questions. Name one medium for rapid reach and one for two-way interaction, and explain why.

ANSWER

TV or SNS for rapid reach; a website with forms/comments or SNS for two-way interaction.

REFLECT & CHALLENGE**Q13**

Which medium do you personally trust most for important news, and why?

ANSWER

Answers vary. Justify using the medium's characteristics and your reason for trusting it.

REFLECT & CHALLENGE**Q14**

Plan a media mix for a school event on a budget that rules out TV. Name one audience the mix might still miss.

ANSWER

Example: website for detail, SNS for spreading, direct messages for specific groups. Identify an audience that may not use those channels.

FILL IN THE BLANK**Q15**

A website cannot be viewed without an (a) connection. Information does not reach users unless they actively (b) it. In actual information dissemination, (c) media are often combined.

ANSWER

- (a) Internet
- (b) access/view
- (c) multiple

MULTIPLE CHOICE**Q16**

Which statement correctly describes media?

- A. Radio conveys video and audio.
- B. Newspapers spread information rapidly.
- C. SNS allows users to post and share.
- D. TV allows two-way interaction.

ANSWER

C.

MULTIPLE CHOICE

Q17

Which statement correctly describes choosing media?

- A. Always use one.
- B. A website is always sufficient.
- C. Choose according to purpose and target audience.
- D. SNS makes TV/websites unnecessary.

ANSWER

C.

4-2 • CHAPTER 4 QUESTION BANK

4-2

Information Design and User Experience for Websites

Worked Example → Try → Exam-Style [6 Marks] → Think as an Engineer → New Context → Reflect & Challenge → Exercise.

TRUE / FALSE

Q1

A persona is a typical user image set up as a concrete person. (True or False?)

ANSWER

True.

TRUE / FALSE

Q2

A wireframe is a final-version design including color and decoration. (True or False?)

ANSWER

False — it is a rough blueprint before visual decoration.

TRUE / FALSE

Q3

Proximity means grouping related information close together. (True or False?)

ANSWER

True.

TRUE / FALSE

Q4

UCD prioritizes the convenience of the creator. (True or False?)

ANSWER

False — UCD designs from the user's perspective.

MATCHING

Q5

Match: (a) highlight important information; (b) repeat style for unity; (c) align positions for order; (d) group related information. Choices: A Contrast, B Repetition, C Alignment, D Proximity.

ANSWER

- (a) → A Contrast
- (b) → B Repetition
- (c) → C Alignment
- (d) → D Proximity

SHORT ANSWER

Q6

What is a typical user image set up as a concrete person called?

ANSWER

Persona.

SHORT ANSWER

Q7

What is the blueprint for placement and priority of information called?

ANSWER

Wireframe.

SHORT ANSWER

Q8

What are the four principles represented by the English initials C-R-A-P?

ANSWER

Contrast, Repetition, Alignment, Proximity.

SHORT ANSWER

Q9

What is the user-perspective design approach and its abbreviation?

ANSWER

User-Centered Design (UCD).

MULTIPLE CHOICE

Q10

Which is NOT appropriate persona information?

- A. User's age/occupation
- B. User's purpose
- C. Website server performance
- D. User's challenges

ANSWER

C.

MATCHING

Q11

Match: large bold headings; group product name/price/photo; align left edges; standardize heading font/color. Choices: A Contrast, B Repetition, C Alignment, D Proximity.

ANSWER

- Large bold headings → A Contrast
- Group product name/price/photo → D Proximity
- Align left edges → C Alignment
- Standardize heading font/color → B Repetition

6-MARK**Q12** 6 marks

Explain why a designer creates a persona and wireframe before choosing colors and fonts. Refer to user needs and stakeholder agreement.

ANSWER

The persona keeps design focused on a concrete user's needs. The wireframe fixes placement and priority before visual preferences, making discussion and agreement with stakeholders easier.

THINK AS AN ENGINEER**Q13**

A community clinic needs a website for opening hours, services and directions. Ask 8–10 people what they would look for first, group answers into user types including a low-vision user or someone with no home Internet, prioritize four homepage blocks, and choose one CRAP principle.

ANSWER

Use evidence rather than taste. Start from the most important/hardest-to-reach need, then choose the CRAP principle that fixes the weakest point.

NEW CONTEXT**Q14**

A school wants parents to find their child's grades quickly on a cluttered results page. Name a persona, one wireframe decision and one CRAP principle, and state what each improves.

ANSWER

Example: parent checking grades; place grades in a high-priority block; use contrast to highlight them or proximity to group related results.

REFLECT & CHALLENGE**Q15**

On a confusing website, which CRAP principle seems most broken, and why?

ANSWER

Answers vary. Explain whether the issue is priority, consistency, order/alignment, or grouping.

REFLECT & CHALLENGE**Q16**

A clinic can fund only one accessibility improvement. Choose one CRAP principle and say what decoration/feature you would give up and why.

ANSWER

Any justified choice is acceptable if it prioritizes the user's needs and explains the trade-off.

FILL IN THE BLANK

Q17

First create a (a) representing a typical user. Next create a (b), a blueprint for placement and priority. Keep the (c) — contrast, repetition, alignment and proximity — in mind.

ANSWER

- (a) Persona
- (b) Wireframe
- (c) CRAP principles

MULTIPLE CHOICE

Q18

Which is NOT part of CRAP?

- A. Contrast
- B. Encryption
- C. Alignment
- D. Proximity

ANSWER

B.

MULTIPLE CHOICE

Q19

Which best describes UCD?

- A. Developer convenience
- B. Visual design first
- C. Begin from what the user needs
- D. Use newest technologies

ANSWER

C.

4-3 • CHAPTER 4 QUESTION BANK

4-3

Methods for Evaluating Websites

Worked Example → Try → Exam-Style [6 Marks] → Think as an Engineer → New Context → Reflect & Challenge → Exercise.

TRUE / FALSE

Q1

Qualitative evaluation uses numerical data to evaluate quantitatively. (True or False?)

ANSWER

False — numerical data is quantitative; qualitative explores users' voices and behavior.

TRUE / FALSE

Q2

PV is the number of times a page has been viewed. (True or False?)

ANSWER

True.

TRUE / FALSE

Q3

The proportion of visits achieving a target action is the bounce rate. (True or False?)

ANSWER

False — it is conversion rate (CVR).

TRUE / FALSE

Q4

Heuristic evaluation uses experts who identify UI problems using rules of thumb. (True or False?)

ANSWER

True.

MATCHING

Q5

Match: (a) page views; (b) visits ending after the first page; (c) visits achieving a target action. Choices: A PV, B Bounce rate, C Conversion rate.

ANSWER

- (a) → A PV
- (b) → B Bounce rate
- (c) → C Conversion rate

SHORT ANSWER

Q6

What is the method that evaluates qualitatively by observing user voices and behavior?

ANSWER

Qualitative evaluation.

SHORT ANSWER

Q7

What is the method that evaluates quantitatively using numerical data?

ANSWER

Quantitative evaluation.

SHORT ANSWER

Q8

What is the two-letter abbreviation for page views?

ANSWER

PV.

SHORT ANSWER

Q9

What is the expert method based on rules of thumb?

ANSWER

Heuristic evaluation.

MULTIPLE CHOICE

Q10

Which are quantitative evaluation methods?

- A. Usability testing
- B. Web analytics
- C. Interviews
- D. A/B testing

ANSWER

B and D.

MULTIPLE CHOICE

Q11

Which describes conversion rate?

- A. Page views
- B. Visits ending after first page
- C. Visits achieving a target action such as purchase or registration
- D. Viewed pages relative to total pages

ANSWER

C.

6-MARK**Q12** 6 marks

Explain why a website team should use both qualitative and quantitative evaluation rather than only one. Refer to what numbers show and what users' voices show.

ANSWER

Quantitative evaluation shows how much is happening; qualitative evaluation explores why. Combining numerical evidence with user voices/behavior and expert checks gives a clearer, more accurate picture.

THINK AS AN ENGINEER**Q13**

An online bookstore has high PV but very low conversion rate. What might that mean? Which method would tell you why? Consider a customer hidden by raw PV data.

ANSWER

Many people arrive but few take the target action. Qualitative testing/interviews can reveal why. Small-screen or first-time users may have difficulties PV alone cannot show.

NEW CONTEXT**Q14**

A school's new website gets many visits, but most people leave after the first page. Name one quantitative indicator, one qualitative method and one likely fix.

ANSWER

Bounce rate; usability testing or interviews; change the page based on the observed cause.

REFLECT & CHALLENGE**Q15**

If you could see only one metric, would you choose PV, bounce rate or CVR, and why?

ANSWER

Answers vary. Choose according to the goal and explain what the metric tells you.

REFLECT & CHALLENGE**Q16**

A page has a high bounce rate. Design a short qualitative test that reveals the cause and say what you would change first.

ANSWER

Observe users performing a task and ask what confused them or what they expected; change the element connected to the observed cause.

FILL IN THE BLANK

Q17

(a) evaluation observes user voices and behavior; (b) evaluation uses numerical data; (c) shows page views; (d) shows visits ending after the first page; (e) shows visits achieving a target action.

ANSWER

- (a) Qualitative
- (b) Quantitative
- (c) PV
- (d) Bounce rate
- (e) Conversion rate/CVR

MATCHING

Q18

Classify: (a) observe users exploring a site; (b) tally access counts/page views; (c) interview users. Choices: A Qualitative, B Quantitative.

ANSWER

- (a) → A Qualitative
- (b) → B Quantitative
- (c) → A Qualitative

MULTIPLE CHOICE

Q19

Which describes heuristic evaluation?

- A. Users explore the site
- B. Numerical data analysis
- C. Experts identify UI problems using rules of thumb
- D. Compare two patterns with data

ANSWER

C.

4-4 • CHAPTER 4 QUESTION BANK

4-4

The Iterative Improvement Process for Websites

Worked Example → Try → Exam-Style [6 Marks] → Think as an Engineer → New Context → Reflect & Challenge → Exercise.

TRUE / FALSE

Q1

PDCA repeats Plan → Do → Check → Act. (True or False?)

ANSWER

True.

TRUE / FALSE

Q2

Design thinking solves problems from the developer's perspective. (True or False?)

ANSWER

False — it works from the user's perspective.

TRUE / FALSE

Q3

A/B testing should change only one element at a time. (True or False?)

ANSWER

True.

TRUE / FALSE

Q4

PDCA is complete after one execution. (True or False?)

ANSWER

False — it is repeated for continuous improvement.

MATCHING

Q5

Match: understand user challenges; clarify problem; generate ideas; create a prototype; have users try it.

Choices: A Empathize, B Define, C Ideate, D Prototype, E Test.

ANSWER

- Understand user challenges → A Empathize
- Clarify problem → B Define
- Generate ideas → C Ideate
- Create a prototype → D Prototype
- Have users try it → E Test

SHORT ANSWER

Q6

What improvement method repeats Plan → Do → Check → Act?

ANSWER

PDCA cycle.

SHORT ANSWER

Q7

What user-perspective approach discovers problems and generates solutions?

ANSWER

Design thinking.

SHORT ANSWER

Q8

What method prepares two patterns and compares them with data?

ANSWER

A/B testing.

SHORT ANSWER

Q9

Name the five design-thinking steps in order.

ANSWER

Empathize → Define → Ideate → Prototype → Test.

MULTIPLE CHOICE

Q10

Which is “Check” in PDCA?

- A. Set goal
- B. Change homepage
- C. Measure bounce rate after the change
- D. Consider another improvement

ANSWER

C.

MULTIPLE CHOICE

Q11

Which is NOT appropriate A/B testing?

- A. Prepare two patterns
- B. Change multiple elements at once
- C. Wait for sufficient data
- D. Show each to actual users

ANSWER

B.

6-MARK**Q12** 6 marks

Explain why a website team improves in repeated PDCA cycles instead of trying to build a perfect site in one go. Refer to measuring results and acting on them.

ANSWER

The first version is a best guess. Real users reveal what works and what does not. PDCA allows a team to plan, make, measure and act on evidence, producing gradual improvement through repetition.

THINK AS AN ENGINEER**Q13**

A school club's event page gets few sign-ups. Lead one measured PDCA loop: choose a starting number, identify one obstacle, plan one change, do it, check again and decide the next step. Choose one element for an A/B test and explain why only that element changes.

ANSWER

Start with a measurable number, identify the biggest obstacle, make one focused change, measure again and use the result for the next cycle. One A/B element is changed so its effect can be identified.

NEW CONTEXT**Q14**

An online shop finds its “buy” button is rarely clicked. Describe one A/B test and one PDCA loop, and name the number that would show the change worked.

ANSWER

Create two button versions changing only one element, show them to actual users and compare clicks/conversion. In PDCA set a goal, make the change, measure it and decide the next step.

REFLECT & CHALLENGE**Q15**

Think of a website you use often. What one small change might help most, and how would you measure whether it worked?

ANSWER

Answers vary. State a specific change and a measurable result.

REFLECT & CHALLENGE**Q16**

You may run one A/B test this month. Choose one element, predict the winning option and state the number that would prove you right or wrong.

ANSWER

Any justified test is acceptable; change one element only and identify a measurable result such as clicks or conversion rate.

FILL IN THE BLANK

Q17

(a) repeated Plan → Do → Check → Act; (b) user-perspective problem/solution approach; (c) two-option data comparison.

ANSWER

- (a) PDCA cycle
- (b) Design thinking
- (c) A/B testing

MATCHING

Q18

Match the five design-thinking descriptions: understand user; clarify problem; generate ideas; create prototype; users try it.

ANSWER

- Understand user → Empathize
- Clarify problem → Define
- Generate ideas → Ideate
- Create prototype → Prototype
- Users try it → Test

REVIEW • CHAPTER 4 QUESTION BANK

Review

Chapter 4 — Final Exam Review

The four lesson chains: 4-1 CHOOSE (purpose + audience → medium) → 4-2 DESIGN (user → persona → wireframe → CRAP → UCD) → 4-3 EVALUATE (quantitative = how much; qualitative = why) → 4-4 IMPROVE (PDCA + design thinking + A/B testing → repeated improvement).

SHORT ANSWER

QRF1

Name the five media compared in 4-1.

ANSWER

TV, radio, newspaper, website, SNS.

SHORT ANSWER

QRF2

Name the four CRAP principles.

ANSWER

Contrast, Repetition, Alignment, Proximity.

SHORT ANSWER

QRF3

What does UCD mean?

ANSWER

User-Centered Design.

SHORT ANSWER

QRF4

Qualitative vs quantitative?

ANSWER

Qualitative explains why; quantitative measures how much using numerical data.

SHORT ANSWER

QRF5

What do PV, bounce rate and CVR mean?

ANSWER

PV = page views; bounce rate = visits ending after the first page; CVR = visits reaching a target action.

SHORT ANSWER

QRF6

What is heuristic evaluation?

ANSWER

Experts identify UI problems using rules of thumb.

SHORT ANSWER

QRF7

What does PDCA stand for?

ANSWER

Plan, Do, Check, Act.

SHORT ANSWER

QRF8

Name the five design-thinking steps.

ANSWER

Empathize, Define, Ideate, Prototype, Test.

SHORT ANSWER

QRF9

Why change only one element in A/B testing?

ANSWER

So the cause of the observed effect can be identified.

SHORT ANSWER

QRF10

Why is website improvement iterative?

ANSWER

The first version is a best guess; real users and measured results show what should change, so teams improve repeatedly.

REFERENCE

QT1

6-Mark Answer Tip

Tip

Do not only list terms. Connect each key concept to the situation in the question and explain the reason. Example: TV → wide reach; website → detailed information; SNS → sharing/word of mouth.

REFERENCE

QM1

Final Memory Map

CHOOSE	→ Media
DESIGN	→ Persona / Wireframe / CRAP / UCD
EVALUATE	→ Qualitative / Quantitative / PV / Bounce / CVR / Heuristics
IMPROVE	→ PDCA / Design Thinking / A/B Testing